

NBQSS 2026 · CHALLENGE 4.1

***AQuaTomo*: A Hardware-Agnostic Quantum State Tomography Suite**

From Haar states and realistic noise to LI, physical MLE, and neural reconstruction

Team members:

Anubhav Kumar Srivastava, ICFO / Ideaded Spain

Ria Patel, NCSU USA

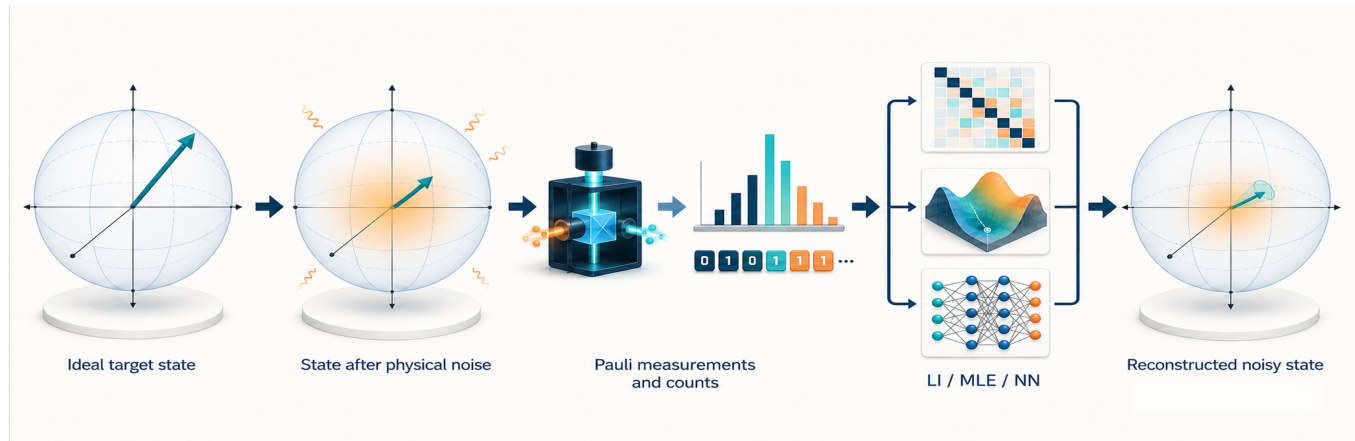
Youngdu Kim, Sungkyunkwan University, South Korea

Github repository:

AQuaTomo: <https://github.com/patelria007/AQuaTomo>

1. What we built—and the contract it obeys

The suite treats tomography as a sequence of independently testable transformations. A state is generated, changed by an explicit quantum channel, measured through Born probabilities and multinomial shots, reconstructed, and assessed against the correct target. The state preparation noise and classical readout noise are independent of each other.



Ideal state → noisy state → classical counts → LI / MLE / NN → physical reconstructed state.

Complete acquisition. For n qubits, every local setting in $\{X, Y, Z\}^n$ is used: 3^n settings and 2^n outcomes per setting.

Backend contract. Numerical kernels discover the array namespace from their inputs. Born-rule linear algebra, eigensolvers, likelihood gradients, and metrics remain on the chosen backend/device. Seeded multinomial sampling and files are explicit host-side control operations because the Array API does not standardize them.

Physical output. MLE uses $\rho = T^\dagger T / \text{Tr}(T^\dagger T)$. The neural model emits a Cholesky-factor parameterization. Both are Hermitian, PSD, and unit trace by construction. Raw LI is retained precisely because its failure is diagnostic as it generates negative eigenvalues on mathematical inversion due to finite dataset.

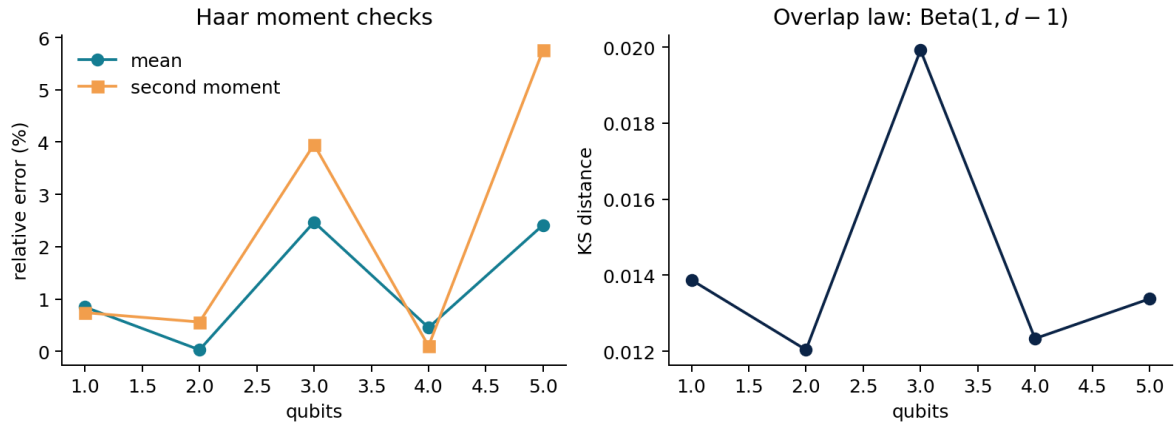
Verification chain. Each stage has an identity or statistical gate: Haar overlap law; analytic channel action; exact-probability LI; likelihood monotonicity; density-matrix invariants; method comparison; scaling cap; and recorded run environment. This ensures the validity of the results shown.

Delivered scientific stack: product/Haar/mixed states; depolarizing, amplitude damping, phase damping, biased Pauli, coherent rotation, and readout-confusion models; finite-shot Pauli data; LI, MLE, direct NN, projection/shrinkage; metrics, CLI, persistence, tests, and reproducibility scripts.

2. Verification I: are the random states genuinely Haar?

For a complex Haar-random ket in dimension $d = 2^n$, draw $z_j \sim \mathcal{N}(0, 1) + i\mathcal{N}(0, 1)$ independently and normalize $|\psi\rangle = z/\|z\|_2$. This normalized isotropic complex Gaussian is unitarily invariant. A strong scalar diagnostic is

$$X = |\langle 0|\psi\rangle|^2 \sim \text{Beta}(1, d-1), \quad \mathbb{E}[X] = \frac{1}{d}, \quad \mathbb{E}[X^2] = \frac{2}{d(d+1)}.$$



n	samples	mean rel. error	second-moment rel. error	KS distance
1	2500	0.84%	0.74%	0.0139
2	2500	0.03%	0.56%	0.0120
3	2500	2.47%	3.96%	0.0199
4	2500	0.45%	0.10%	0.0123
5	2500	2.41%	5.76%	0.0134

Verification passed. Across 12,500 independent states, every KS distance is below the pre-declared 0.05 threshold. Maximum trace and purity errors are 6.7×10^{-16} and 1.3×10^{-15} . This verifies both individual-state physicality and the expected ensemble distribution; checking purity alone would not establish Haar randomness.

Measurement identity. Exact probabilities are $p_{s,b} = \langle b|U_s^\dagger \rho U_s|b\rangle$. With probability-weighted “counts”, Pauli LI recovers independent Haar targets for $n = 1, \dots, 5$ with Frobenius error $\leq 4.95 \times 10^{-16}$. Finite-shot counts sum exactly to the requested shots for every setting.

3. Verification II: does each noise model do what its name claims?

Real devices need more than isotropic depolarization. The suite now separates quantum-state channels from classical assignment error and provides analytic one-qubit checks before any reconstruction benchmark.

Model	First-order meaning	experimental	Analytic check	Result
Global/local depolarizing	isotropic stochastic mixing / effective twirled error		Bloch vector shrinks by $1 - p$	$p = .2 : r_x = 0.800$
Amplitude damping	energy relaxation, finite T_1		$ 1\rangle$ population becomes $1 - \gamma$	$\gamma = .3 : \rho_{11} = 0.700$
Phase damping	loss of phase coherence, finite T_2 contribution		off-diagonal scales by $1 - \lambda$	$\lambda = .25 : \rho_{01} = 0.375$
Biased Pauli	anisotropic stochastic/control error		CPTP weights sum to one	$\text{trace} = 1 + 2.2 \times 10^{-16}$
Coherent rotation	systematic over/under rotation		unitary channel preserves purity	$\text{purity} = 1.000$
Readout confusion	asymmetric assignment/SPAM outcomes	assignment in classical	$C_{ij} = P(i_{\text{obs}} j_{\text{ideal}})$ is column stochastic	exact one/two-qubit probabilities

Verification passed. Every analytic identity agrees within 2.3×10^{-16} ; all tested quantum-channel outputs are Hermitian, PSD, and trace one. Readout confusion acts only on outcome probabilities and therefore does not falsely claim to modify the latent quantum state.

Platform-oriented model sets.

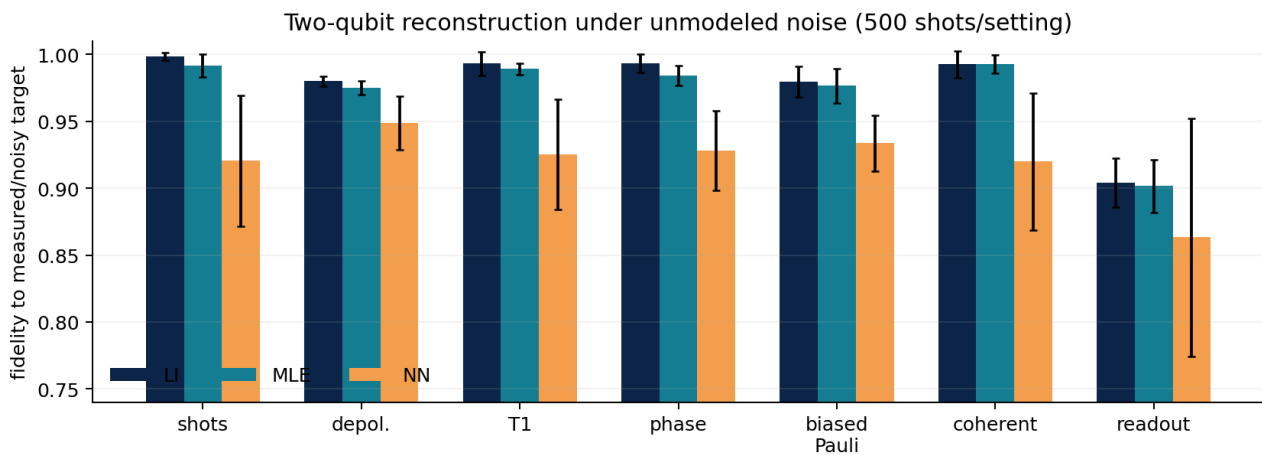
- Superconducting: amplitude/phase damping, coherent control, crosstalk, leakage, readout assignment.
- Trapped ions: dephasing, coherent over-rotations, correlated crosstalk, preparation/measurement error.
- Photonics: loss/erasure, detector inefficiency and dark counts, mode mismatch, phase drift.
- Neutral atoms: atom loss/leakage, dephasing, blockade/crosstalk, SPAM.

Next fidelity upgrade. Fit channel/readout parameters from calibration data, permit qubit- and time-dependent rates, then test the calibrated likelihood on dates/devices withheld from training. Correlated channels should use explicit Kraus/process or latent calibration models; a single depolarizing number is a baseline, not a hardware model.

Correct target. For a state channel, reconstruction is scored against the post-noise state; fidelity from ideal to noisy target separately measures damage. For readout error, the latent state is unchanged, so an uncorrected estimator's degradation measures model mismatch.

4. Verification III: reconstruction works, but, with evident failures

Estimator	Definition	Strength	Failure mode
Linear inversion	$\hat{\rho}_{LI} = 2^{-n} \sum_P \langle \hat{P} \rangle P$	fast, transparent, exact without sampling noise	finite shots produce negative eigenvalues; fidelity is not a valid state metric until physicalized
Maximum likelihood	minimize multinomial NLL with $\rho = T^\dagger T / \text{Tr}(T^\dagger T)$	physical, shot-aware, auditable objective	repeated full-data passes; sensitive to likelihood mismatch
Direct neural net-work	frequencies $\rightarrow$ MLP $\rightarrow$ Cholesky parameters $\rightarrow \rho$	physical, very fast amortized inference	training cost, dimension/design specificity, distribution shift



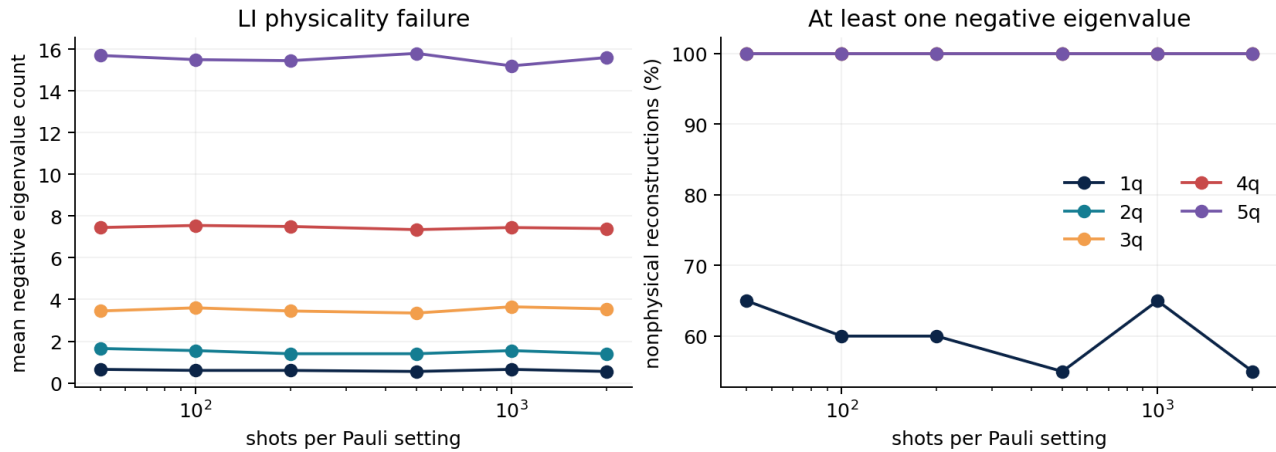
Two-qubit validation used 500 shots per local-Pauli setting, eight independent Haar targets per noise case, seven noise/readout cases, and 60 MLE iterations. The neural network was the independently trained 64–64 Cholesky-output model (600 training states; 60 held-out clean test states), not the attention architecture from the separate paper.

Method	mean target fidelity across seven cases	physical fraction	mean inference time*
LI	0.977 [†]	0.125	0.00036 s
MLE	0.973	1.000	0.068 s
NN	0.920	1.000	0.00022 s

[†]LI's value is a diagnostic only when its matrix is nonphysical. *Local CPU smoke-test timing, not a backend comparison.

Verification passed, but NN does not win this regime. MLE and NN are physical in every noise case. The compact NN is fastest at inference but less accurate, especially for unmodeled readout error (mean fidelity 0.863). This is evidence for calibration-aware/OOD training—not evidence that a larger black-box network automatically solves scaling.

5. Why LI remains nonphysical as shots increase



The study used Haar-*pure* targets, 20 independent states at each point, $n = 1, \dots, 5$, and 50–2000 shots per setting. For $n \geq 2$, every LI matrix had at least one eigenvalue below -10^{-10} at every tested shot count. At five qubits LI carried roughly 15–16 negative eigenvalues on average.

This apparently flat count is expected. A pure density matrix is on the boundary of the PSD cone: its $d - 1$ true zero eigenvalues receive positive and negative $O(N^{-1/2})$ perturbations. More shots reduce the *magnitude* of the negative tail—for five qubits the mean minimum eigenvalue improves from -0.256 at 50 shots to -0.041 at 2000—but the sign count need not vanish at finite N .

Interpretation. “More shots” makes raw LI statistically closer, not automatically physical. Always report minimum eigenvalue or PSD fraction alongside fidelity. For a state estimate, use constrained MLE or an exact projection; for a transparent diagnostic, retain raw LI and show its spectrum.

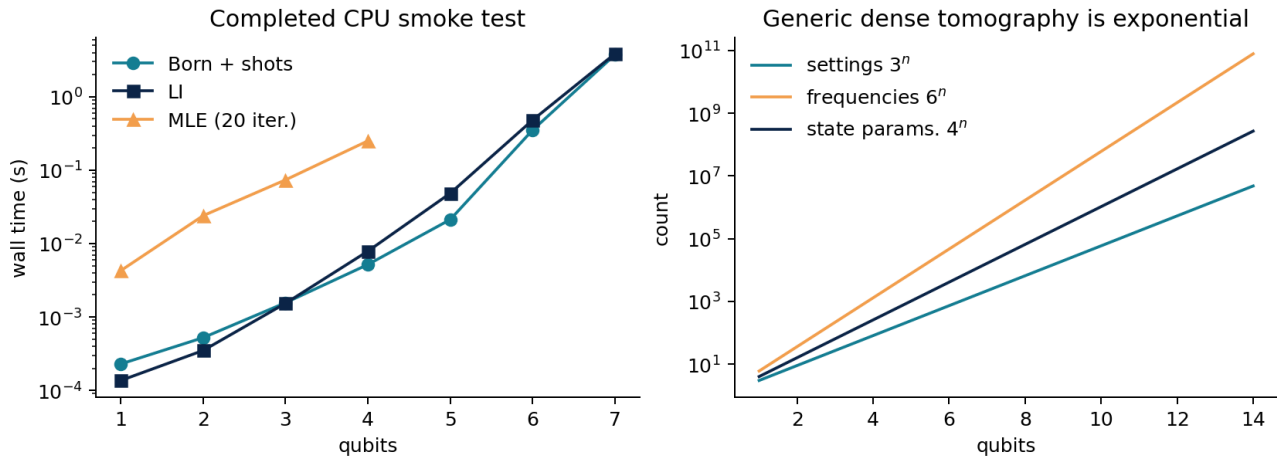
Projection baseline. Hermitize, diagonalize, project the eigenvalue vector onto the probability simplex, and reconstruct. This is the closest PSD trace-one matrix in Frobenius norm.

Rank is a hypothesis. Rank-one truncation can be excellent for truly pure states and strongly biased for mixed states. Choose rank using held-out likelihood, information criteria, or a physically motivated model.

Shot splitting. The suite splits observed counts by multivariate-hypergeometric sampling, preserving totals without resimulating the unknown state. Denoising strength can then be selected on an independent validation partition.

Coverage still matters. Point fidelity alone cannot certify a reconstruction procedure. Bootstrap or likelihood-ratio regions and observable coverage should be added before experimental scientific claims.

6. Maximum implemented size and the observed scaling cliff



The complete dense NumPy pipeline was increased until the next size failed an explicit cap on the same machine. Seven qubits completed: 2,187 settings, 279,936 frequencies, 16,384 Pauli coefficients, 200 shots/setting, 3.73 s for Born probabilities plus sampling, and 3.85 s for LI. The projected LI fidelity for that single smoke target was 0.889. The eight-qubit probe requires 6,561 settings and 1,679,616 frequencies; acquisition alone did not finish within 45 s, so reconstruction was not started in that capped run.

n	3^n settings	6^n frequencies	4^n parameters	copies at 500/setting	observed status
5	243	7,776	1,024	121,500	complete
6	729	46,656	4,096	364,500	complete
7	2,187	279,936	16,384	1,093,500	complete
8	6,561	1,679,616	65,536	3,280,500	acquisition > 45 s cap
12	531,441	2,176,782,336	16,777,216	265,720,500	theoretical only; frequency vector $\approx$ 16.2 GiB

What fails first. The dense array for one ρ is not yet the main bottleneck at eight qubits. Enumerating every setting, rotating a dense state, storing 6^n frequencies, and then revisiting them across MLE iterations dominate. A GPU accelerates kernels; it does not change the exponents or the number of experimental copies.

Scope of “maximum”. Seven qubits is the largest configuration actually completed by this reference implementation under the stated local run; it is not a universal hardware limit. The capped eight-qubit failure is reproducible with `tools/probe_scaling_limit.py`.

7. Classical shadows and learned estimators beyond dense QST

The scalable choice is often a smaller output. A dense MLP or attention model can improve amortized inference and learn repeatable device bias, but its complete-Pauli input still has 6^n frequencies and a generic physical output still represents 4^n parameters. Local-Pauli classical shadows instead retain randomized measurement records for estimating selected observables without constructing a dense state.

For a weight- k Pauli string, the unbiased single-shadow estimator contributes only when every supported random basis matches, with probability 3^{-k} . One shadow data set can answer many low-weight queries chosen after acquisition. The implementation follows Huang, Kueng, and Preskill and supports empirical standard errors plus median-of-means aggregation.

The matched-budget study used two-qubit product and Haar states, 20 replicates, four observables (ZI, IZ, ZZ, XX), and 90–4500 state copies. Complete-QST methods used nine settings; the direct split baseline divided the same collection budget among four separately targeted settings.

Method at 4500 copies	aggregate observable RMSE	interpretation
Rank-one MLE	0.0191	best here because the pure-state rank prior is correct
Direct targets, split budget	0.0264	strong for four observables fixed before acquisition
Classical shadows	0.0345	one reusable data set; queries may be selected afterward
Linear inversion	0.0439	unconstrained full density matrix
Neural network	0.1671	physical and fast, but fixed two-qubit training plateaued

Observed trade-off. Shadows improved on raw LI and the fixed neural network while avoiding a full-state output. Rank-one MLE remained strongest because its structural assumption exactly matched the simulated pure states. For one known observable, direct measurement remains the honest default; shadows become attractive when a larger low-weight collection must share data or be selected post hoc.

When NN can win.

- The measurement design and device noise repeat across many states.
- The architecture encodes locality, symmetry, low rank, or tensor structure.
- Training cost is amortized over many reconstructions.
- Device/date/noise holdouts and uncertainty checks are passed.

When it should not be trusted.

- A new device/date/noise level lies outside the training support.
- Dense attention is advertised as many-body scalability.
- Only in-distribution average fidelity is reported.
- The network learns the state prior rather than the measurement inverse.

The shot-grid study also produced a surprise: product states did not consistently require fewer shots than Haar states to reach median 0.99 fidelity. The generic rank-one MLE encoded purity, not tensor-product structure. A product ansatz and many more replicates are needed before claiming a family-level separation.

8. Reproduce on a laptop and on a GPU cluster

Environment that produced the reported CPU data. One local process on a MacBook Air (MacBookAir10,1), Apple M1 with 8 CPU cores (4 performance + 4 efficiency), 16 GB unified memory, and 7-core integrated GPU; macOS 26.5.2 arm64; Python 3.12.9; NumPy 1.26.4 linked to OpenBLAS 0.3.21. Thread variables were unset; scheduler variables were unset; no cluster node was used. JAX and CuPy were not installed, and the Apple GPU was not used. Timings in this report are therefore NumPy/CPU smoke tests only.

Accelerator execution. The CuPy/CUDA and JAX benchmark commands below were subsequently run on one cluster node with one NVIDIA RTX 4060 Ti, PCIe 4, 8 GB GPU memory, and “sm 8.7”. This confirms execution of both backend paths for one and two qubits at 100 and 1000 shots. No stdout/CSV timing results were supplied, so the CPU figures and timing statements elsewhere in this report remain the local NumPy measurements; no GPU speedup is inferred.

Laptop CPU

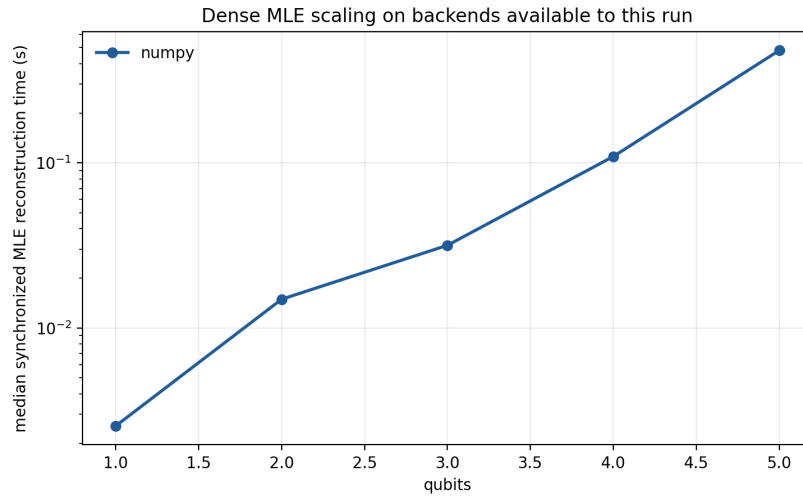
```
cd nbqst_suite
python -m venv .venv && source .venv/bin/activate
python -m pip install -e '[array-api,test]' matplotlib
PYTHONPATH=src python -m pytest -q
PYTHONPATH=src python tools/run_verification_study.py
PYTHONPATH=src python tools/probe_scaling_limit.py -qubits 8 -timeout 45
python examples/neural_comparison.py
```

Confirmed one-node CuPy/CUDA and JAX smoke run

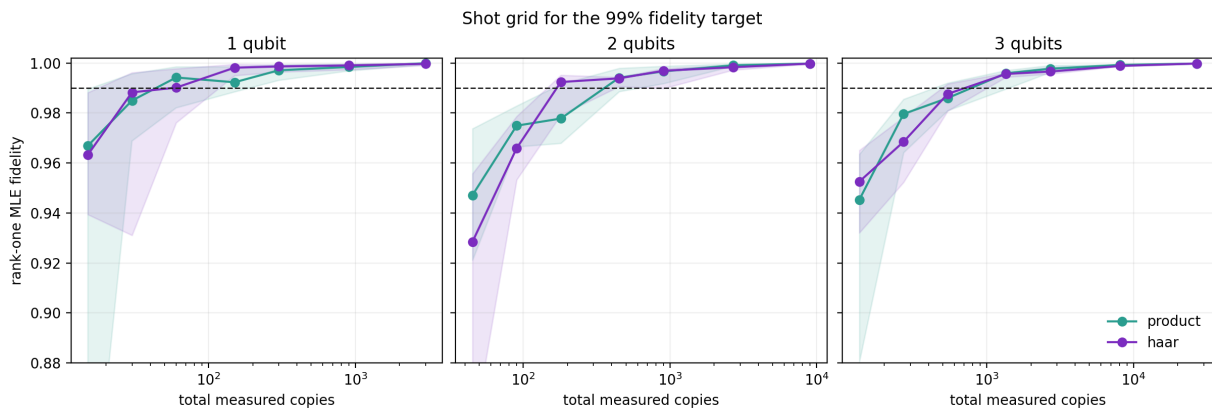
```
1 node; 1 × NVIDIA RTX 4060 Ti; PCIe 4; 8 GB; sm 8.7
nbqst benchmark -backend cupy -qubits 1 2 -shots 100 1000
nbqst benchmark -backend jax -qubits 1 2 -shots 100 1000
```

9. Additional analytical findings and classical shadows

These studies isolate three questions that are easy to conflate: computational scaling, the experimental copies needed for high-fidelity state recovery, and the copies needed to estimate selected observables. All plotted CPU timings are local NumPy measurements on the Apple M1 configuration reported in Section 8.

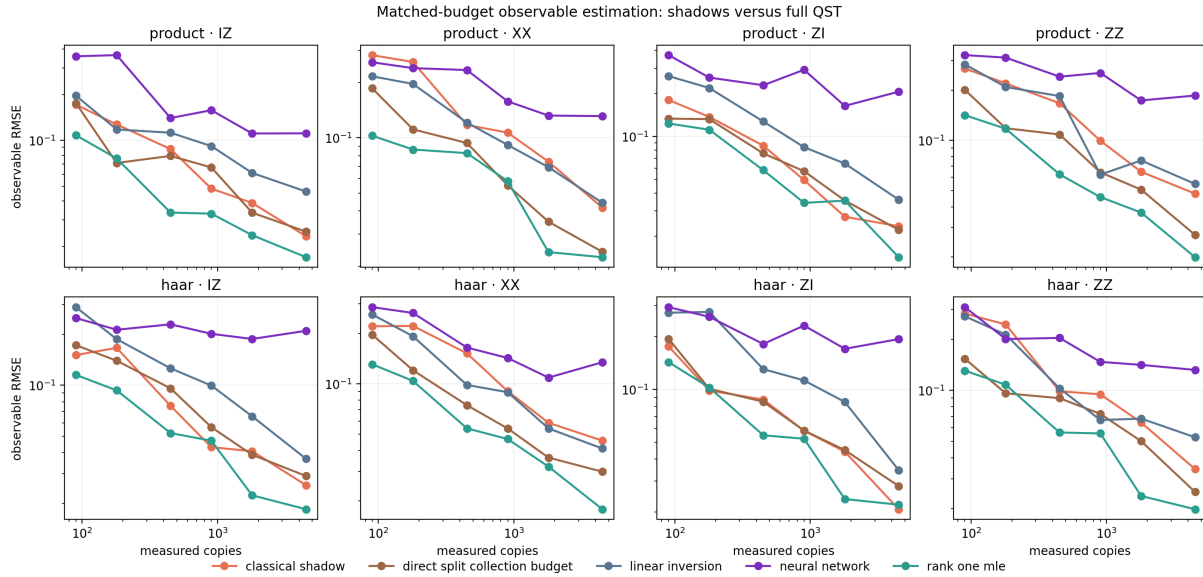


Dense MLE timing on the local CPU. Median reconstruction time rose from 0.00255 s at one qubit to 0.48050 s at five qubits, a factor of about 188.



Small-system shot threshold for median fidelity ≥ 0.99 under the tested rank-one MLE protocol. Results use simulated one- to three-qubit pure states.

The shot-grid result does not support a general claim that product states are always easier than Haar states. At two qubits, the observed threshold was 50 shots per setting for product states and 20 for Haar states; at three qubits both required 50. The study has limited replicates, and the estimator uses a pure-state prior rather than an explicit product-state ansatz, so these values are diagnostic rather than asymptotic sample-complexity laws.



Matched collection-budget comparison for four two-qubit Pauli observables (ZI , IZ , ZZ , XX), averaged across Haar and product pure states with 20 replicates.

At 4,500 copies, aggregate observable RMSE was 0.0191 for rank-one MLE, 0.0264 for direct measurement with the budget split among four targets, 0.0345 for classical shadows, 0.0439 for LI, and 0.1671 for the fixed NN. Shadows therefore improved on unconstrained LI and reused one randomized data set for post-hoc queries, but they did not beat a correctly specified pure-state MLE in this small matched-prior experiment. Direct measurement remained best suited to a small set of observables known before acquisition; measuring each target with the full budget achieved RMSE 0.0133 but required four times the total collection.

Combined conclusion. GPUs can reduce kernel time, but dense full-state QST still inherits exponential measurement, storage, and output costs. Classical shadows change the objective from recovering every entry of ρ to predicting chosen observables, which is the more scalable question when a complete density matrix is unnecessary.

10. Conclusions, limitations, and next decision

What is established.

- The Gaussian-normalization generator passes Haar overlap moments and a distributional KS gate through five qubits.
- Six state/readout noise mechanisms pass analytic action and physicality checks.
- Exact complete-Pauli LI recovers Haar states through five qubits to $< 5 \times 10^{-16}$ Frobenius error.
- MLE and NN outputs remain physical under every tested noise case; raw LI often does not.
- Seven-qubit dense generation + LI completes locally.
- The current small direct NN is fast but less accurate than MLE; it is a baseline, not a scaling claim.
- Shadows beat raw LI for the tested observable collection; rank-one MLE remains best under a correct purity prior.

What is not yet established.

- No claim about real device performance: the current experiments are simulated.
- No calibrated correlated noise, leakage, loss-aware POVM, drift, or uncertainty intervals yet.
- CuPy/CUDA and JAX execution is confirmed on one RTX 4060 Ti node, but no synchronized accelerator timing or speedup claim is established.
- No evidence that dense attention overcomes exponential full-state output.
- Eight replicates per noise case validate integration, not publication-level confidence.

Recommended next decision. Use MLE as the dense physical reference, retain LI as an auditable failure-revealing baseline, and evaluate neural denoising only against held-out device/noise conditions. Beyond roughly this dense regime, choose the target first: low-rank/tensor-network state reconstruction for structured states, or classical shadows when only observables are needed.

Reproducibility artifacts. The source tree contains Array-API kernels, an object-oriented pipeline facade, standalone neural and classical-shadow functions, 23 tests, an executed quick-start notebook, raw CSV/JSON results, environment capture, and the scaling probe. Every numeric statement above is traceable to those files.

11. Failed attempts, surprises, and what we would change

Failed attempts and limitations.

- Raw LI remained nonphysical at finite shots, especially for pure states on the PSD boundary. More shots reduced the negative eigenvalue magnitudes but did not reliably eliminate their count.
- The first dense scaling path became impractical at eight qubits: complete acquisition exceeded the 45 s cap before reconstruction began. GPU execution cannot remove the 6^n frequency table or 4^n state description.
- The compact Cholesky-output NN plateaued on Haar states and did not exploit additional shots as effectively as likelihood-based reconstruction. Its fixed training distribution, limited Haar coverage, parameter-space loss, and lack of uncertainty calibration made it a useful baseline rather than a successful denoiser.
- The accelerator smoke tests did not produce an auditable performance comparison.

What surprised us.

- The number of negative LI eigenvalues can stay nearly flat while their magnitudes shrink, because finite-shot perturbations split the many zero eigenvalues of a pure state around zero.
- Product states did not consistently reach 0.99 fidelity with fewer shots than Haar states. A purity prior alone does not encode tensor-product structure.
- Classical shadows were not universally best: they were valuable for reusable post-hoc observable estimation, while direct measurement won for a few preselected observables and rank-one MLE won when its state prior was exactly correct.
- Neural inference was fast and always physical, yet this did not translate into better accuracy. Physical parameterization and low latency are necessary but not sufficient for robust learned tomography.

What we would do differently with more time.

- Collect calibrated hardware data across devices and dates; learn or fit readout, relaxation, dephasing, coherent, leakage, and correlated-noise parameters, then evaluate on held-out devices and drift periods.
- Benchmark NumPy, CuPy, and JAX from identical frozen count bundles with synchronized warm-ups, repeated trials, precision controls, and automatically archived scheduler/hardware manifests.
- Replace the generic NN loss with the multinomial likelihood or a fidelity-aligned objective, encode purity, locality, and symmetry, train separate mixed- and pure-state models, and report calibration and out-of-distribution uncertainty.
- Move beyond dense matrices using low-rank factors, matrix-product states, adaptive measurements, compressed sensing, or shadows according to the target state and desired observables.
- Increase independent replicates, attach bootstrap confidence intervals, and pre-register comparison budgets so claims about shot efficiency and scaling are statistically defensible.

Final perspective. Reconstruction, denoising, and noise mitigation are different tasks. LI and standard MLE estimate the state that generated the observed data; recovering an ideal pre-noise state requires a calibrated forward noise model or additional error-mitigation assumptions. Future work should focus on denoising the reconstructed states.

Selected references

- [1] NBQSS 2026, “Building a Hardware-Agnostic Quantum State Tomography Suite,” Challenge 4.1 (provided brief).
- [2] A. Macarone Palmieri et al., “Enhancing quantum state tomography via resource-efficient attention-based neural networks,” *Physical Review Research* 6, 033248 (2024), [doi:10.1103/PhysRevResearch.6.033248](https://doi.org/10.1103/PhysRevResearch.6.033248).
- [3] D. Koutný, L. Motka, Z. Hradil, J. Řeháček, and L. L. Sánchez-Soto, “Neural-network quantum state tomography,” [arXiv:2206.06736](https://arxiv.org/abs/2206.06736) (2022).
- [4] J. A. Smolin, J. M. Gambetta, and G. Smith, “Efficient method for computing the maximum-likelihood quantum state from measurements with additive Gaussian noise,” *Physical Review Letters* 108, 070502 (2012).
- [5] D. Gross et al., “Quantum state tomography via compressed sensing,” *Physical Review Letters* 105, 150401 (2010).
- [6] H.-Y. Huang, R. Kueng, and J. Preskill, “Predicting many properties of a quantum system from very few measurements,” *Nature Physics* 16, 1050–1057 (2020).

AI disclosure. OpenAI Codex was utilized for software architecture, verification tests, and schematic figures. Everything was checked and cross-verified by unit tests, exact-state identities, negative-log-likelihood monotonicity, cross-method comparisons, and document reviews by the team members. The scientific review, the compilation and execution of codes, data analysis, and the figure-generation were done by the team members.